

1A.

S is set of all pos ints, so $S = \{n \in \mathbb{N} \mid 6|n\}$

$\rightarrow \{6k \mid k \in \mathbb{N}\}$

$\mathbb{N}$ is countably infinite and S is infinite subset of $\mathbb{N}$ /

An infinite subset of countably infinite set is countably infinite.

Therefore, S is countably infinite.

1B.

$A = \{0, 1, 2, 3\} \times \mathbb{Z}^+$

$A_1 \cup A_2 \cup A_3 \cup A_4$ where...

$A_1 = \{0\} \times \mathbb{Z}^+ \mid A_2 = \{1\} \times \mathbb{Z}^+ \mid A_3 = \{2\} \times \mathbb{Z}^+ \mid A_4 = \{3\} \times \mathbb{Z}^+$

$\mathbb{Z}^+$ is countably infinite by definition.

A_1 is are all countably infinite because $f: A_1 \rightarrow \mathbb{Z}^+$ by $f((0, n)) = n$ is a bijection.

A_2, A_3, A_4 are also countably infinite.

Countable union of countable set is countable $\rightarrow A_1 \cup A_2 \cup A_3 \cup A_4$ is countable

Since A is infinite, A is countably infinite.

2.

Use Cantor's diagonalization argument to show the set is larger in cardinality than the set of natural numbers $\mathbb{N}$, which proves it uncountable.

Assume for contradiction that F is countable, where $f_1, f_2, f_3, \dots$ where each f is a function from $\mathbb{N}$ to $\mathbb{N}$.

New function $g: \mathbb{N} \rightarrow \mathbb{N}$ is constructed, where $g(n) = f_n(n) + 1$, guaranteeing that no value of g is equivalent to f for any n.

G is a function from $\mathbb{N}$ to $\mathbb{N}$ and should be in the list of functions F.

However by construction, g differs from f_n at every position, since $g(n) \neq f_n(n)$ for all n.

This means g cannot be a function in list, which contradicts the original assumption that list contained all functions from $\mathbb{N}$ to $\mathbb{N}$.

Therefore, the set of functions from $\mathbb{N}$ to $\mathbb{N}$ is uncountable.

3A.

S is set of all real numbers with decimal representation consisting of all 9's.

$S_n \in S$, where n = number of 9's that S_n has.

Then $S_n = \{s_1s_2\dots s_n, s_1s_2\dots s_{(n-1)}s_n, \dots, s_1s_2s_3\dots s_n\}$

$|S_n| = n$ and each S_n is countable.

Now $S = \bigcup A_n$ where $n \in \mathbb{N}$

Therefore, S is the countable set and thus S is countable.

3B.

Let S be set of all real numbers with decimal representation consisting of 8's and 9's.

Let T be set of all real numbers with decimal representation in the form of $3a_1a_2\dots$

where $a_i \in \{8, 9\}$.

As seen above, T is a subset of S .

If possible, let set T be countable.

$T = \{r_1, r_2, r_3, \dots\}$ where $r_n = 3a_{n1}, 3a_{n2}, \dots a_{ni}$

$a_{nj} \in \{8, 9\}$ and $\forall j \in \mathbb{N}$

Define $x = 3$ and $x_1, x_2, x_3\dots$ where

$x_i = 8$ iff $a_{ii} = 9$ and $x_i = 9$ iff $a_{ii} = 8$

Then $x \neq r_i$ but $x \in T$ is a contradiction.

Therefore T is uncountable and since T is a subset of S , S is also uncountable.

4A.

$A = [4, 5]$ is uncountable.

$B = [5, 6]$ is uncountable.

$A \cap B = [4, 5] \cap [5, 6] = \{5\}$

$\{5\}$ is finite.

A and B are uncountable because there are many real numbers between the values mentioned in the sets. The intersection between the two sets leaves just one finite number (5), satisfying the requirement for this example.

4B.

$A = (-4, -1) \cup [1, \infty)$

$B = (-\infty, -4] \cup [-1, 1] \cup \{2, 3, 4, 5\dots\}$

A and B as shown above are both uncountable.

But $A \cap B = \{1, 2, 3, 4, 5\dots\}$

This intersection represents the set of natural numbers, which is countably infinite.

A and B are uncountable because the portions of the sets between any two finite values will be uncountably infinite, along with any other infinite interval. The only countable interval in the two sets is the $\{2, 3, 4, 5\dots\}$ in set B , but everything else is uncountable. When the two sets are intersected, what results is $\{1, 2, 3, 4, 5\dots\}$ which is another way to portray a set of natural numbers. Therefore, it is countably infinite as $\mathbb{N}$ is countably infinite.

4C.

$A = \mathbb{R}$ (all real numbers)

$B = \mathbb{R}^+$ (all real numbers greater than or equal to 0)

A and B as shown above are both uncountable.

$A \cap B = \mathbb{R}^+$

$\mathbb{R}^+$ is still uncountably infinite.

Sets A and B are uncountable because they stretch onwards towards negative or positive infinity and include a vast, uncountable amount of elements. When the two sets are intersected, the results basically equal what set B already equates to. Since set B is a set of real numbers greater than zero, it is still uncountably infinite.